

HTTP/2 vs HTTP/3

Core Protocol Comparison

Feature	HTTP/2	HTTP/3
Release Year	2015	2022 (RFC 9114)
Transport Protocol	TCP	QUIC (UDP-based)
Encryption	Optional TLS 1.2+	Mandatory TLS 1.3
Multiplexing	Yes (stream-based)	Yes (improved stream-based)
Header Compression	HPACK	QPACK
Server Push	Yes	Yes (improved implementation)
Connection Migration	No	Yes
Fallback Mechanism	HTTP/1.1	HTTP/2 or HTTP/1.1

Performance Characteristics

Performance Aspect	HTTP/2	HTTP/3
Connection Setup	2015	2022 (RFC 9114)
Head-of-Line Blocking	TCP	QUIC (UDP-based)
Packet Loss Recovery	Optional TLS 1.2+	Mandatory TLS 1.3
Latency on Poor Networks	Yes (stream-based)	Yes (improved stream-based)
Mobile Performance	HPACK	QPACK
Network Switching	Yes	Yes (improved implementation)

Technical Features

HTTP/2 Features

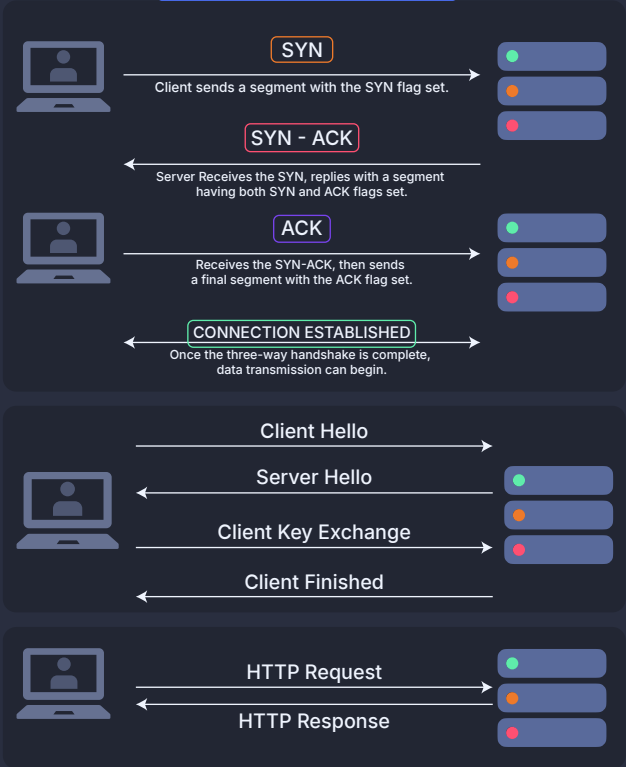
HTTP/2 Protocol Stack
HTTP Semantics
Stream Multiplexing & Flow Control
HPACK Header Compression
TLS 1.2+ (optional but recommended)
TCP
IP

HTTP/3 Features

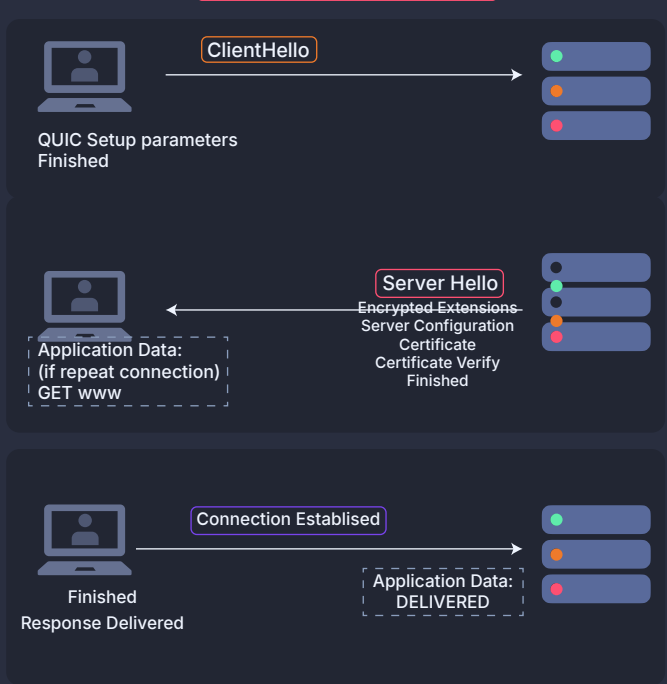
HTTP/3 Protocol Stack
HTTP Semantics
Stream Multiplexing & Flow Control
QPACK Header Compression
QUIC (includes TLS 1.3)
UDP
IP

Connection Establishment

TCP Handshake

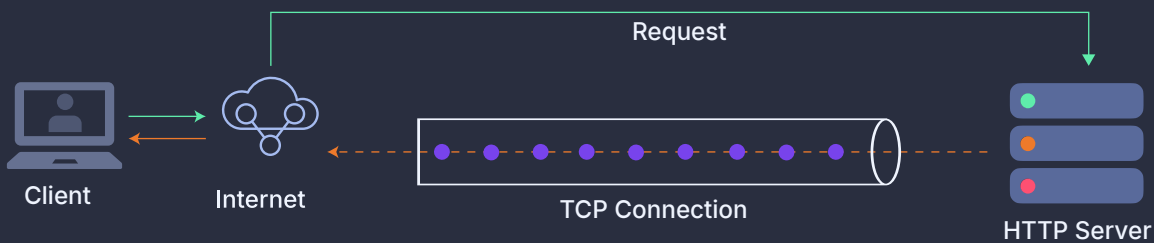


QUIC Handshake

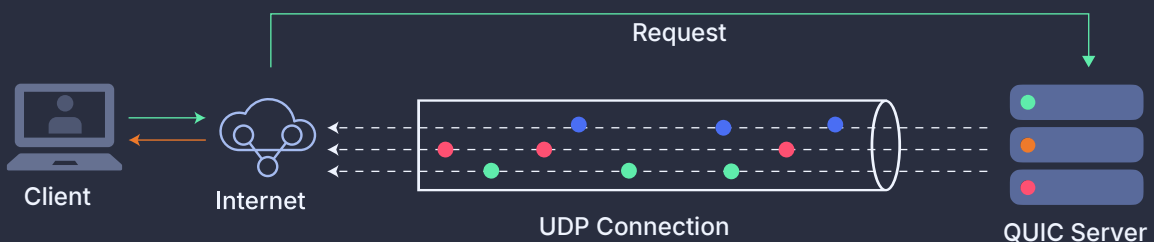


MULTIPLEXING

HTTP/2



HTTP/3 (QUIC)



HTTP/2 Implementation

Server Requirements

- Modern web server (Apache 2.4.17+, Nginx 1.9.5+, IIS 10+)
- TLS certificate (de facto requirement)
- Proper TCP tuning
- HTTP/2 module enabled

Client Support

- All modern browsers (Chrome, Firefox, Safari, Edge)
- Some older mobile browsers limited or no support

Deployment Config

```
# Example Nginx HTTP/2 configuration
server {
    listen 443 ssl http2;
    ssl_certificate
    /path/to/cert.pem;
    ssl_certificate_key
    /path/to/key.pem;
    # Other settings...
}
```

HTTP/3 Implementation

Server Requirements

- Updated web server with HTTP/3 module (Nginx, Caddy, LiteSpeed)
- TLS 1.3 support
- UDP port 443 open and routable
- QUIC library implementation

Client Support

- Chrome 87+, Edge 87+, Firefox 88+, Safari 14+
- Mobile support improving but still evolving

Deployment Config

```
# Example Nginx HTTP/3 configuration
server {
    listen 443 ssl http2;
    listen 443 quic;
    ssl_certificate /path/to/cert.pem;
    ssl_certificate_key /path/to/key.pem;
    add_header Alt-Svc 'h3=":443"; ma=86400';
    # Other settings...
}
```